

DANDONG, China

WMO: 544970

Lat: 40.0317N

Lon: 124.3275E

Elev: 14

StdP: 101.16

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-14.2	-28.1	0.3	-12.5	-25.5	0.4	-10.8	9.4	-7.1	8.6	-7.2	2.9	340	0.433

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	30.2	24.0	28.9	23.3	27.7	22.7	26.1	28.2	25.2	27.1	24.4	26.2	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.5	20.8	27.2	24.7	19.7	26.3	23.9	18.8	25.6	81.3	28.3	77.6	27.3	74.2	26.2	30.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.4	DB	-18.6	32.6	2.7	1.5	-20.5	33.6	-22.1	34.5	-23.6	35.3	-25.6	36.4	(4)
(5)			WB	-19.3	27.1	2.6	1.3	-21.2	28.0	-22.7	28.7	-24.1	29.5	-26.0	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-6.7	-3.1	2.6	9.6	15.6	20.4	23.8	24.1	19.4	12.1	3.2	-4.6
	DBStd	11.27	4.00	4.04	3.44	3.28	2.54	2.16	1.85	1.99	2.67	3.75	4.89	4.24
	HDD10.0	1844	517	368	229	45	1	0	0	0	0	23	208	453
	HDD18.3	3599	776	602	487	261	90	6	0	0	19	196	453	711
	CDD10.0	1760	0	0	1	34	174	313	427	438	283	87	4	0
	CDD18.3	475	0	0	0	0	6	69	168	179	52	1	0	0
	CDH23.3	2576	0	0	0	2	47	298	873	1142	209	5	0	0
	CDH26.7	511	0	0	0	0	5	51	188	251	16	0	0	0
Wind	WSAvg	2.7	3.1	3.0	3.1	3.0	2.6	2.2	2.1	2.2	2.4	2.7	3.1	3.1
Precipitation	PrecAvg	968	9	11	18	53	66	112	286	232	93	44	29	12
	PrecMax	1472	60	41	54	145	144	267	654	508	195	170	61	32
	PrecMin	670	1	0	2	8	10	21	98	39	17	0	4	1
	PrecStd	212	11	10	13	35	37	63	144	128	47	37	17	9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.3	8.9	14.8	22.8	27.2	30.2	32.4	32.3	28.5	23.5	16.5	7.3
		MCWB	1.4	3.7	7.2	13.5	17.0	21.1	26.1	26.1	22.2	17.2	13.2	5.4
	2%	DB	2.4	6.1	11.7	19.5	24.5	28.1	30.3	30.4	26.8	21.1	14.2	4.8
		MCWB	-0.2	1.8	5.6	11.5	15.7	19.7	24.4	24.5	20.5	15.6	11.2	2.5
	5%	DB	0.8	4.3	9.8	17.3	22.7	26.4	28.7	29.2	25.6	19.4	12.5	3.0
		MCWB	-1.7	1.4	4.7	10.3	15.0	19.6	23.7	23.9	19.7	14.5	9.6	0.8
	10%	DB	-0.7	2.8	8.1	15.5	21.1	25.0	27.3	28.0	24.4	17.9	10.5	1.5
		MCWB	-3.1	0.1	4.0	9.6	14.5	19.6	23.3	23.6	19.1	13.7	7.6	-0.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	4.6	9.2	15.4	19.1	23.0	27.6	27.7	24.2	18.9	14.0	6.0
		MCDB	3.7	6.6	12.7	19.4	23.9	27.8	30.7	29.8	26.7	21.5	15.3	7.1
	2%	WB	0.3	3.0	7.1	13.1	17.6	21.9	26.1	26.3	22.7	17.2	12.1	2.8
		MCDB	1.9	5.0	9.6	16.7	21.7	25.0	28.5	28.2	25.0	19.5	13.6	4.2
	5%	WB	-1.4	1.8	5.9	11.9	16.6	21.2	25.1	25.6	21.7	16.0	10.1	1.2
		MCDB	0.5	4.0	8.4	15.8	20.2	24.2	27.0	27.5	23.9	18.2	11.9	2.7
	10%	WB	-3.0	0.4	4.7	10.7	15.8	20.5	24.2	24.9	20.7	14.7	8.2	-0.4
		MCDB	-1.0	2.5	7.5	14.0	19.3	23.4	25.9	26.7	22.9	16.9	10.1	1.2
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.7	9.2	9.0	7.2	5.6	6.9	9.2	9.8	8.1	7.5
		MCDBR	9.1	9.7	10.8	12.1	12.0	10.5	8.5	8.8	10.1	10.8	9.6	8.4
		MCWBR	7.0	6.7	6.3	6.0	5.3	4.3	3.6	3.9	4.7	6.1	7.0	6.8
	5% WB	MCDBR	8.6	8.6	9.4	10.0	9.6	7.9	6.2	6.2	7.9	8.9	8.5	7.8
MCWBR		7.2	6.6	6.4	6.4	5.8	5.0	3.8	3.2	3.5	4.6	6.0	7.2	7.1
Clear-Sky Solar Irradiance	taub	0.402	0.478	0.538	0.624	0.652	0.734	0.710	0.604	0.530	0.556	0.489	0.415	
	taud	1.962	1.813	1.735	1.614	1.611	1.526	1.614	1.820	1.918	1.762	1.853	1.951	
	Ebn at Noon	706	699	707	679	674	620	630	688	710	616	601	650	
	Edh at Noon	129	174	210	252	258	282	256	203	173	182	145	122	
All-Sky Solar Radiation	RadAvg	2.41	3.17	4.33	5.00	5.59	4.94	4.19	4.51	4.43	3.32	2.31	2.05	
	RadStd	0.16	0.23	0.31	0.40	0.46	0.60	0.44	0.52	0.36	0.21	0.24	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page